Sahaj Bhadja

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Summary

Software engineer with 2+ years building **Python**, **Java**, and **React** systems across AI applications, data pipelines, microservices, and cloud platforms. Experienced in integration, prototyping, automated testing, defect resolution, and production delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Software Engineering, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, OOP, Software Engineering, Data Structures and Algorithms

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected a multi-tenant **Python** platform with Azure SQL and Blob Storage for 500+ users, optimizing data models, indexed queries, and access paths to reduce dashboard latency from 1.2 seconds to 350 milliseconds.
- Built cloud data-integration workflows processing 5,000+ monthly jobs with incremental loads, reconciliation checks, error handling, automated monitoring, repeatable recovery, and CI/CD-backed releases, achieving 99%+ reliability across production environments.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Built **React** interfaces and Python Flask services handling 10,000+ daily requests, integrating multimodal AI, secure file ingestion, REST APIs, and responsive workflows while maintaining stable performance during peak load.
- Developed a multithreaded data-processing pipeline for 1,000+ files using validation checks, fault-tolerant execution, progress monitoring, defect recovery, and user-visible status feedback, reducing end-to-end processing turnaround by 60%.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built a Python **AI pipeline** processing 100,000+ autonomous-driving trajectory logs, improving long-horizon planning by 18% through reproducible experiments, automated validation, performance tracing, and repeatable structured model configuration.
- Implemented spatial-temporal modeling that combined HD-map geometry with vehicle trajectories through multi-head attention, improving closed-loop stability 15% while supporting systematic debugging and comparative analysis across complex scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Engineered enterprise integration software with automated builds and VM-based validation across Windows, Linux, and macOS, maintaining 20+ unit and integration test suites through Jenkins, PyTest, and Mockito.
- Developed object-oriented microservices and AWS data pipelines using Glue, S3, SQL, caching, and pagination, improving response times 30%, processing efficiency 68%, and maintainability across integrated platforms with reliable operations.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed **React**, TypeScript, and Node.js features across eight microservices, integrating WebSockets and REST APIs to synchronize more than two million records across responsive customer and operational workflows.
- Automated testing and cloud delivery with Jenkins, Docker, Jest, Mocha, Selenium, Sentry, and Azure monitoring, cutting deployment time 50% while improving release validation and production observability across environments.

Skills

Languages: Python, Java, JavaScript, TypeScript, SQL, C++, Bash, HTML, CSS
Applications & APIs: React.js, Next.js, Flask, FastAPI, Node.js, Express.js, REST APIs, WebSockets, microservices, responsive UI, system integration
Data & AI: PostgreSQL, MySQL, MongoDB, Azure SQL, AWS Glue, S3, RDS, ETL pipelines, data validation, PyTorch, Transformers, LLM systems
Cloud & Quality: AWS, Azure, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, PyTest, Mockito, Jest, Selenium, Sentry, logging, monitoring
Engineering Practices: Git, Agile, SDLC, OOP, system design, prototyping, unit testing, integration testing, defect resolution, code reviews, technical documentation

Projects

Big-Brain | Live | *Python, LLaMA-2, FAISS, PostgreSQL, GCP*
- Built an AI document platform processing 50,000+ records with semantic search, summarization, secure access, and incremental indexing, reducing retrieval latency 40% through optimized embedding and similarity-search pipelines.

Chat2SQL | Code | *Java, TypeScript, Docker, Maven, SQL*
- Built a cross-platform Java and TypeScript application supporting 10+ databases, schema-aware query generation, structured validation, automated testing, and Docker-based deployment across Windows, macOS, and Linux development environments.

PiXMod | Live | *Next.js, PyTorch, Transformers, MongoDB*
- Built a cloud AI application using Next.js, Transformers, and MongoDB for segmentation, retrieval, and generative editing, processing 1,000+ assets through responsive workflows, scalable metadata storage, and deployment.